Computer Networks (CL3001)

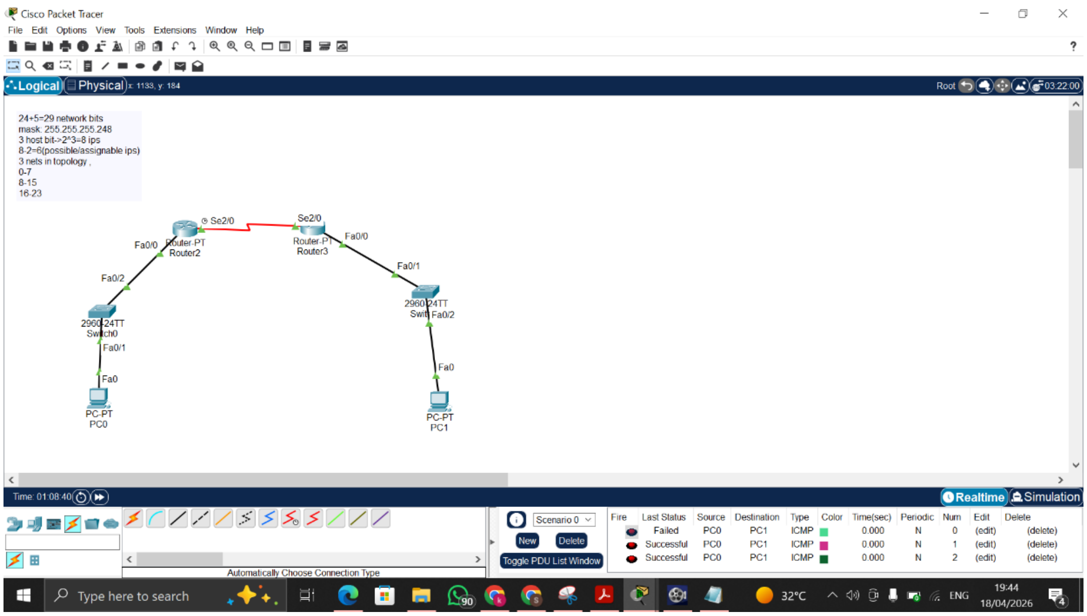
CN LAB 10

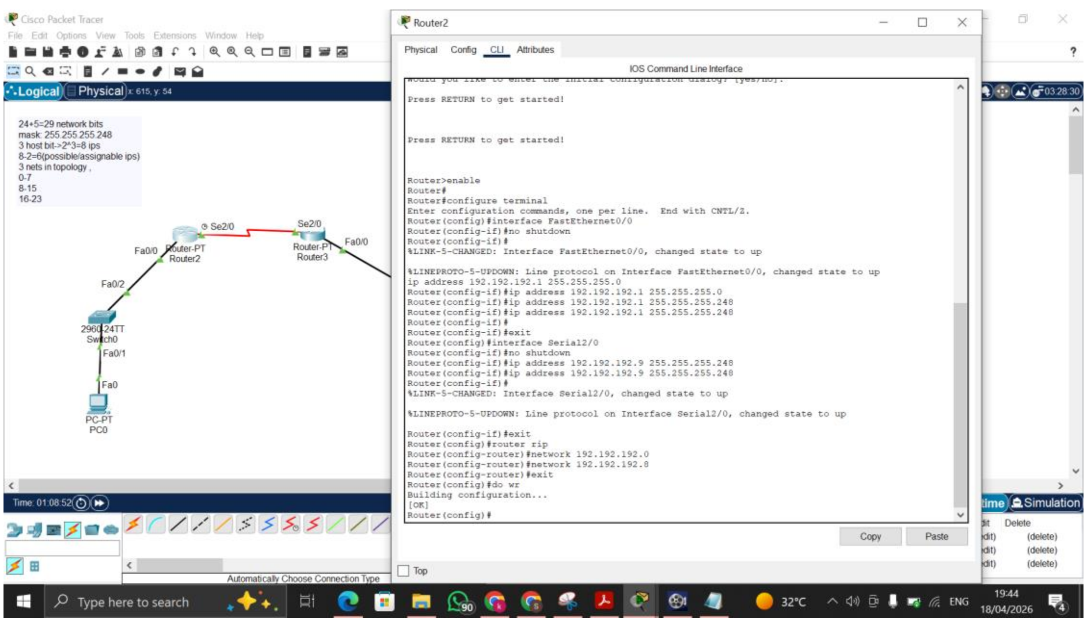
Submitted by:
Syeda Sara Ali
23K-0070
BS-AI (6A)

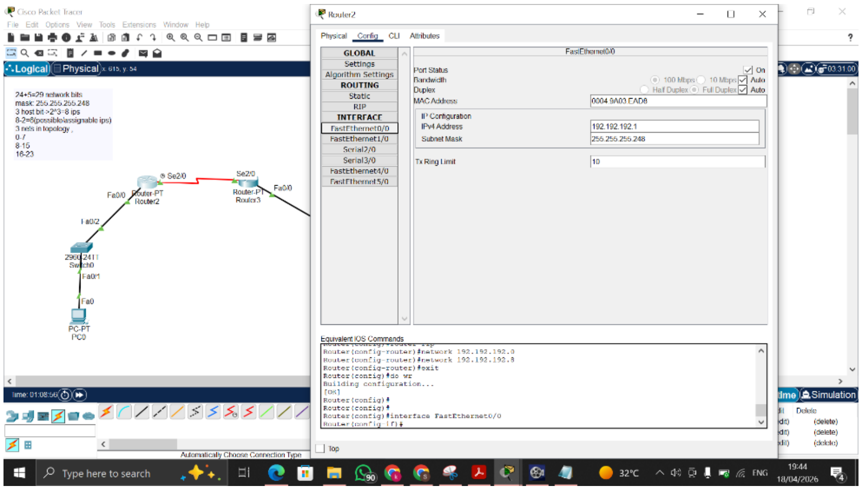
Submitted to:
Sir Sameer Faisal

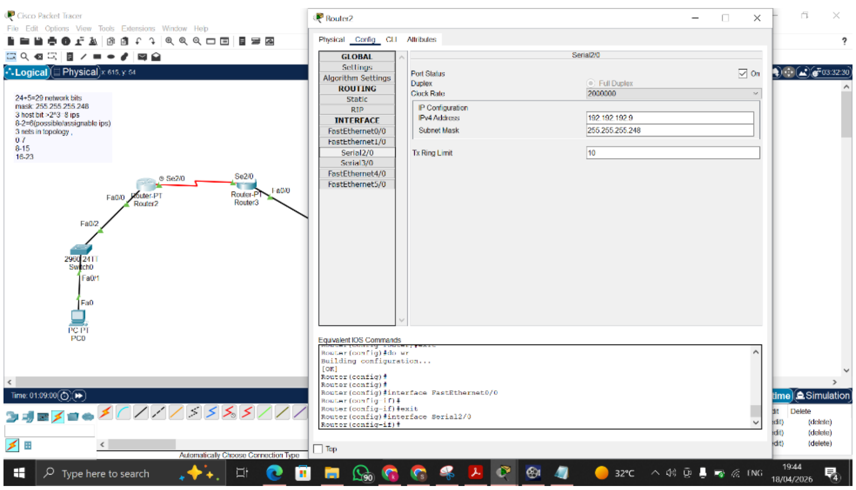
Lab Tasks:

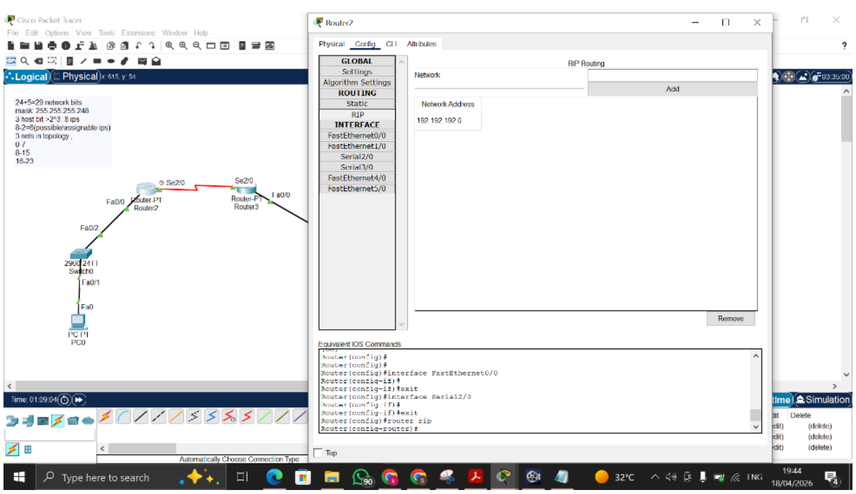
1. *Implement the given example in the manual (Figure 6) on Cisco Packet Tracer, but use the network address 192.192.192.0/29 using RIP.*

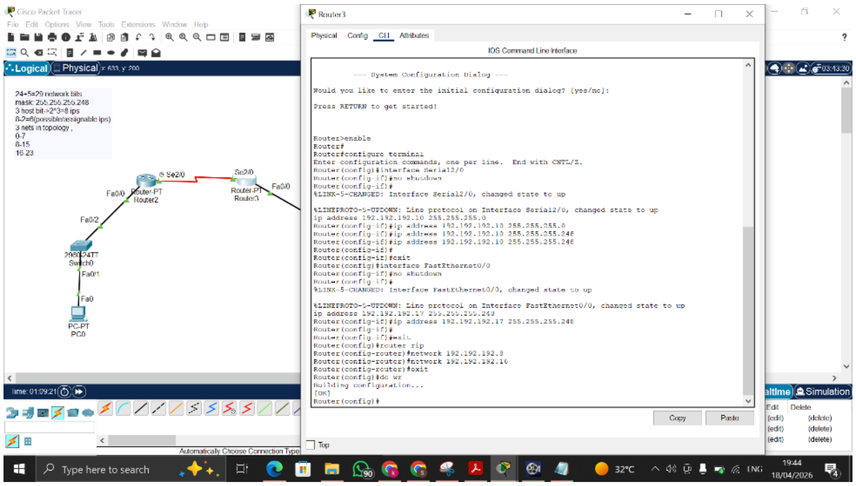


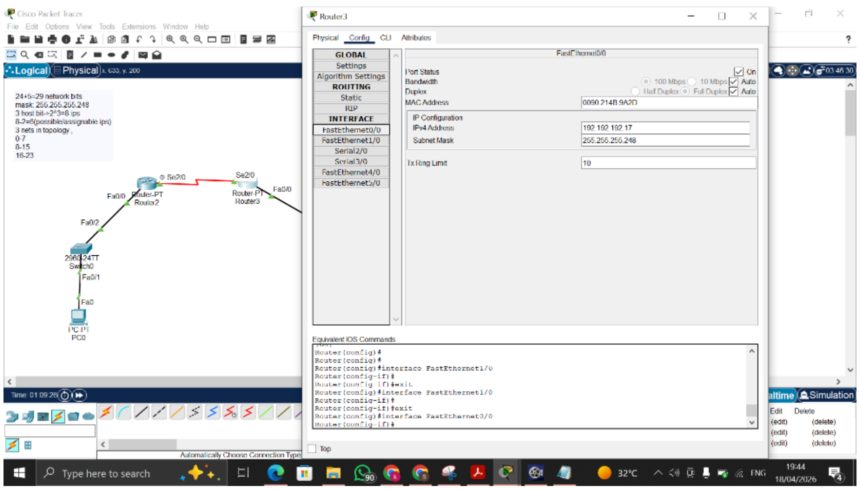


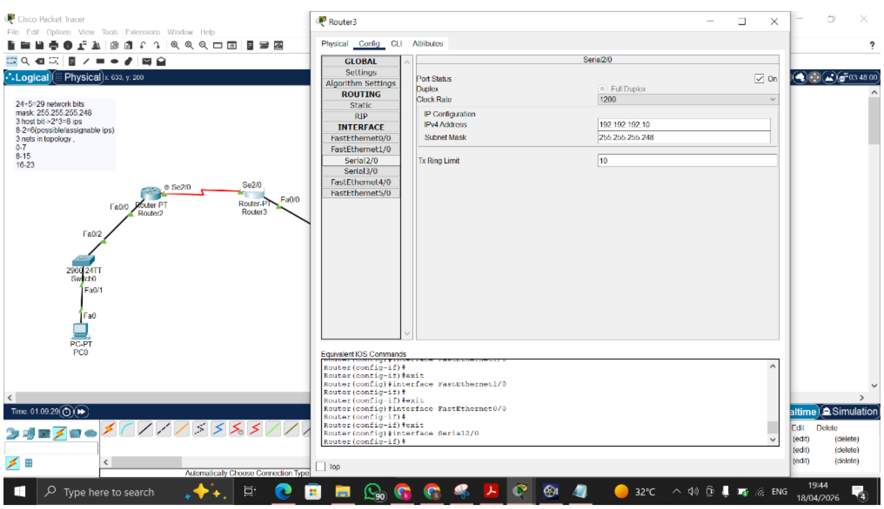

Router2
FastEthernet0/0
192.192.192.1
255.255.255.248

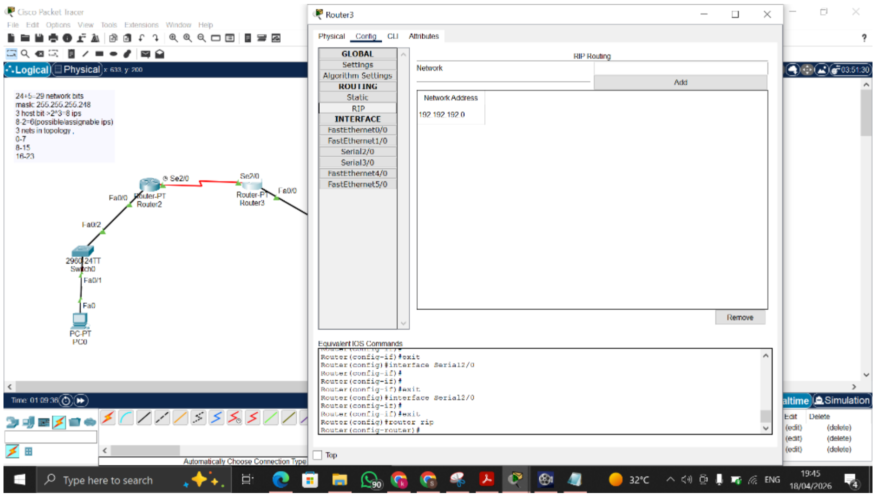

Router2
Serial2/0
2000000
192.192.192.9
255.255.255.248

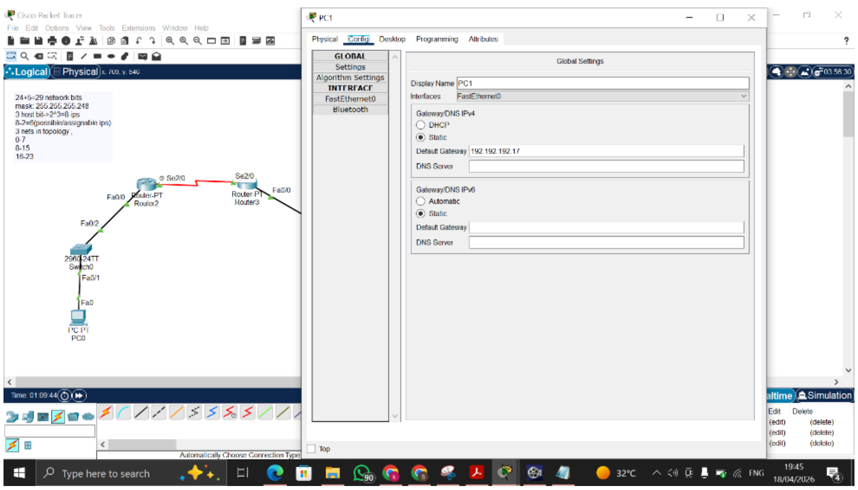

Router2
RIP Routing
Network
Add
Network Address
192.192.192.0
Remove

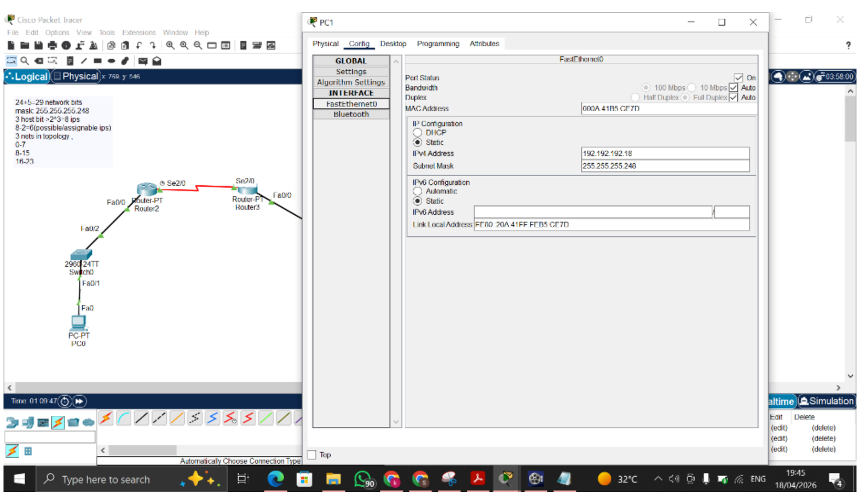

Cisco Packet Tracer
24+5=29 network bits
mask: 255.255.255.248
3 host bit->2^3=8 ips
8-2=6(possible/assignable ips)
3 nets in topology ,
0-7
8-15
16-23
Se2/0
Fa0/0
Router-PT
Router2
Router3
Fa0/2
2960-24TT
Switch0
Fa0/1
Fa0
PC-PT
PC0
Router3
Physical
Config
CLI
Attributes
IOS Command Line Interface
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 192.192.192.10 255.255.255.0
Router(config-if)#ip address 192.192.192.10 255.255.255.0
Router(config-if)#ip address 192.192.192.10 255.255.255.248
Router(config-if)#ip address 192.192.192.10 255.255.255.248
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 192.192.192.17 255.255.255.248
Router(config-if)#ip address 192.192.192.17 255.255.255.248
Router(config-if)#
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 192.192.192.8
Router(config-router)#network 192.192.192.16
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
Copy
Paste
Top
Type here to search
32°C
ENG
19:44
18/04/2026

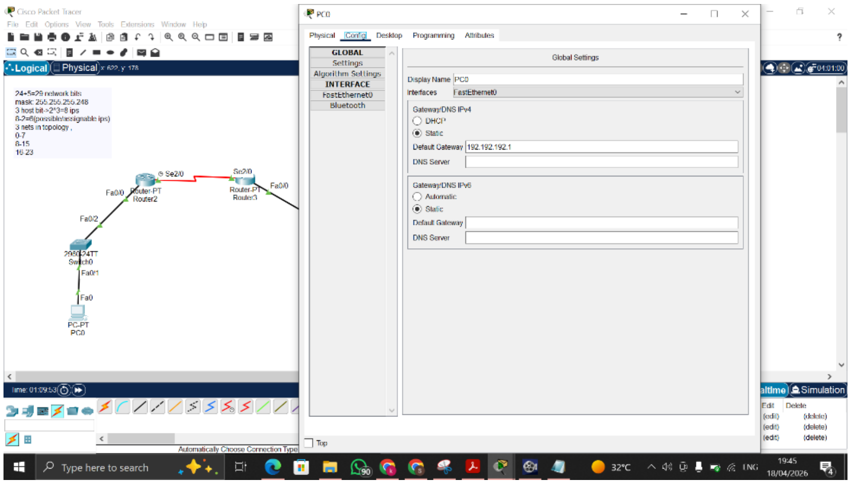

Cisco Packet Tracer
24+5=29 network bits
mask: 255.255.255.248
3 host bit->2^3=8 ips
8-2=6(possible/assignable ips)
3 nets in topology ,
0-7
8-15
16-23
Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
FastEthernet0/0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.214B.9A2D
IP Configuration
IPv4 Address
192.192.192.17
Subnet Mask
255.255.255.248
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-if)#exit
Router(config)#interface FastEthernet1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Type here to search
32°C
ENG
19:44
18/04/2026

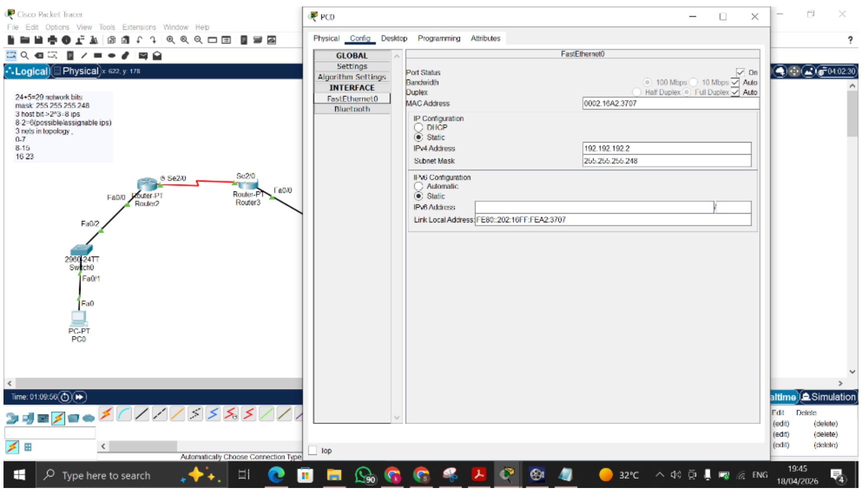

Cisco Packet Tracer
24+5=29 network bits
mask: 255.255.255.248
3 host bit->2^3=8 ips
8-2=6(possible/assignable ips)
3 nets in topology ,
0-7
8-15
16-23
Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial2/0
Port Status
On
Duplex
Full Duplex
Clock Rate
1200
IP Configuration
IPv4 Address
192.192.192.10
Subnet Mask
255.255.255.248
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet1/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
Type here to search
32°C
ENG
19:44
18/04/2026

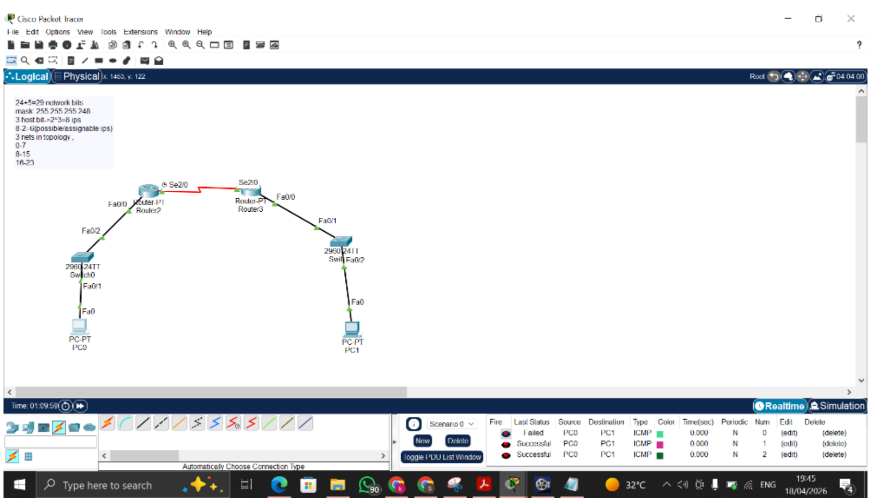

Router3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
RIP Routing
Network
Add
Network Address
192.192.192.0
Remove
Equivalent IOS Commands
Top
24+5=29 network bits
mask: 255.255.255.248
3 host bit->2^3=8 ips
8-2=6(possible/assignable ips)
3 nets in topology ,
0-7
8-15
16-23
Router-PT
Router2
Router3
2960-24TT
Switch0
PC-PT
PC0
Type here to search
32°C
ENG
19:45
18/04/2026


PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC1
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.192.192.17
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Type here to search
32°C
ENG
19:45
18/04/2026


PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
IP Configuration
DHCP
Static
IPv4 Address 192.192.192.18
Subnet Mask 255.255.255.240
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Top
Type here to search
32°C
ENG
19:45
18/04/2026


Cisco Packet Tracer
PC0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC0
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.192.192.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
24+5=29 network bits
mask: 255.255.255.248
3 host bit->2^3=8 ips
8-2=6(possible/assignable ips)
3 nets in topology ,
0-7
8-15
16-23
Se2/0
Fa0/0
Router-PT
Router2
Router3
Fa0/2
2960-24TT
Switch0
Fa0/1
Fa0
PC-PT
PC0
Realtime
Simulation
Type here to search
32°C
ENG
19:45
18/04/2026


Cisco Packet Tracer
PC0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address 0002.16A2.3707
IP Configuration
DHCP
Static
IPv4 Address 192.192.192.2
Subnet Mask 255.255.255.248
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::202:16FF:FEA2:3707
Top
24+5=29 network bits
mask: 255.255.255.248
3 host bit->2^3=8 ips
8-2=6(possible/assignable ips)
3 nets in topology ,
0-7
8-15
16-23
Type here to search
32°C
ENG
19:45
18/04/2026


Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
24+5=29 network bits
mask: 255.255.255.248
3 host bit->2^3=8 ips
8-2=6(possible/assignable ips)
3 nets in topology ,
0-7
8-15
16-23
Se2/0
Fa0/0
Router-PT
Router2
Router3
Fa0/1
Fa0/2
2960-24TT
Switch0
Switch1
Fa0
PC-PT
PC0
PC1
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Failed
Successful
PC0
PC1
ICMP
0.000
N
(edit)
(delete)
Type here to search
32°C
ENG
19:45
18/04/2026

2. *Let`s consider an example of subnetting for ABC Company. There are 3 departments, i.e. Finance, HR and Development. You must perform subnetting for the allocation of the given requirement:*
   32 PCs for Development
   27 PCs for Finance
   10 PCs for HR
   The network address for the given scenario is 200.16.100.0/24. Only show calculation.

CN-Lab 10

Ques 2:-

3 → deparments → 3 network

Dev → 32
Fin → 27 → Hosts
HR → 10

net addr → 200.16.100.0/24

max hosts → 32
min networks → 3

32+2 → 34 required
So 64 → host bits
2 bits to borrow

2^2 = 4 networks
$\boxed{256 \div 4 = 64}$

255.255.255.11000000 = 2^7+2^6 = 192
255.255.255.192 → subnet mask

net addr → 200.16.100.0/26
Dev = 0 – 63
Fin = 64 – 127 → Host ranges
HR = 128 – 191

3. *Let`s consider an example of subnetting for FAST NUCES. There are 3 departments, i.e. CS, EE and BBA. You must perform subnetting for the allocation of the given requirement:*
30 PCs for CS
31 PCs for EE
20 PCs for BBA
The network address for the given scenario is 195.168.10.0/24. Implement it on Cisco Packet Tracer and apply RIPv2.

Ques 3:-

3 departments → 3 networks
CS → 30
EE → 31
BBA → 20

net addr → 195.168.10.0/24.

max hosts → 31
min networks → 3

31+2 = 33 required
so 64 host bits
2 bit to borrow

$2^2 = 4$ net
256/4 = 64

255.255.255.192 → subnet mask

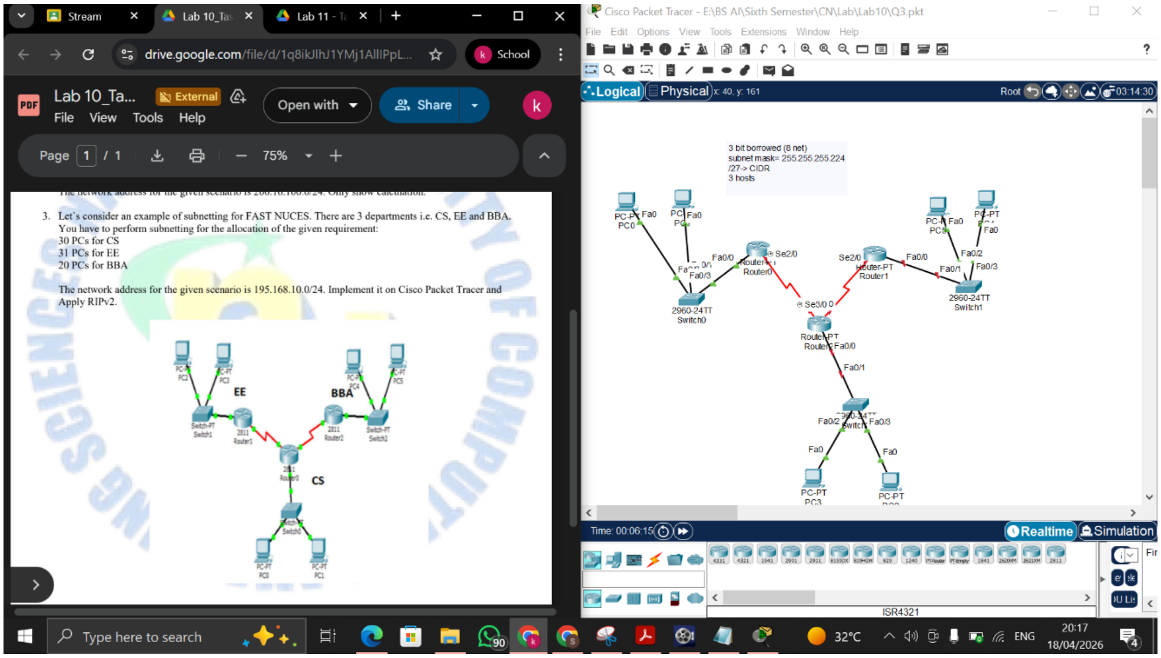
net addr → 195.168.10.0/26.

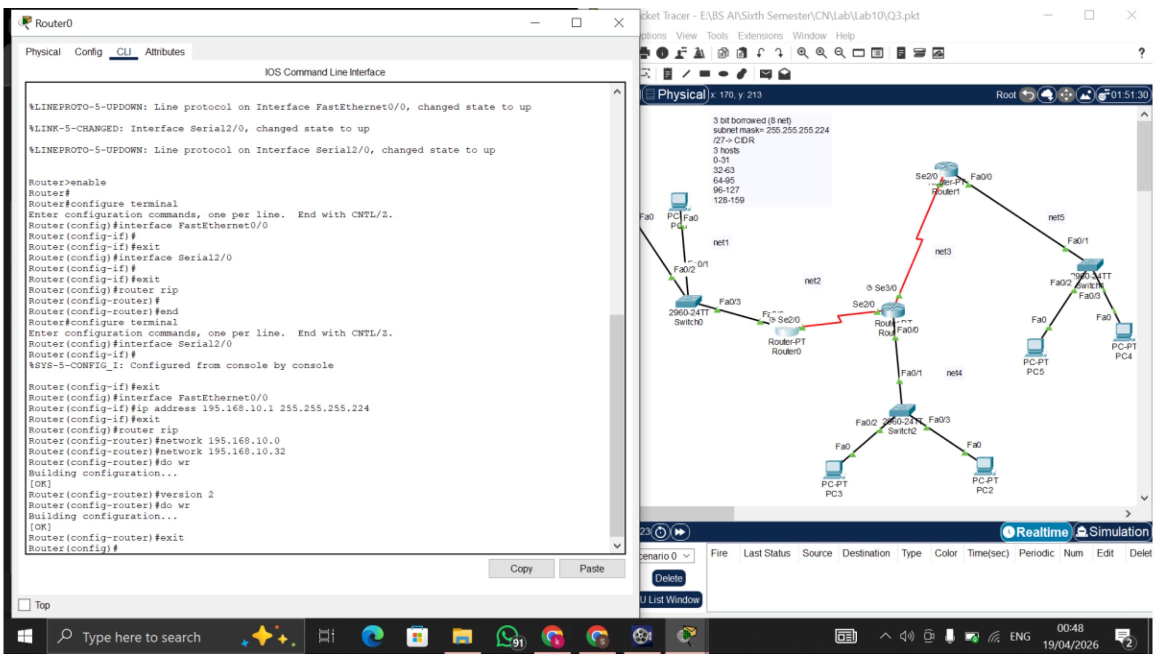
CS → 0 – 63
EE → 64 – 127
BBA → 128 – 191

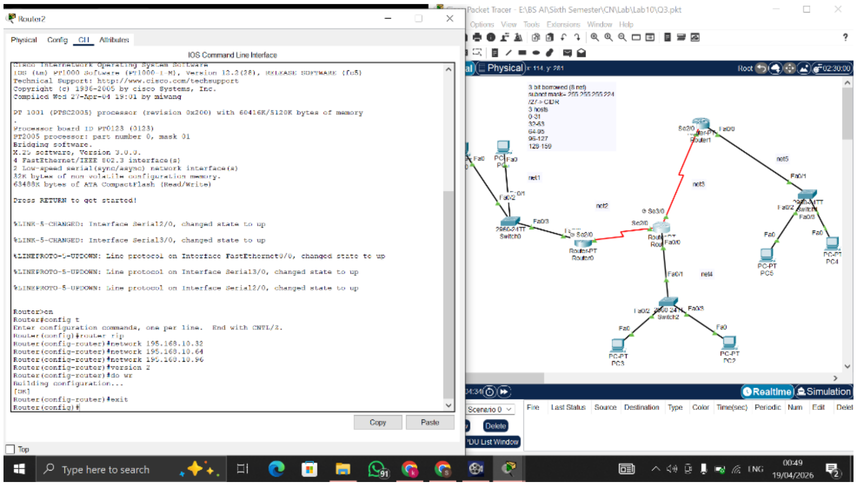
→ Host rangs

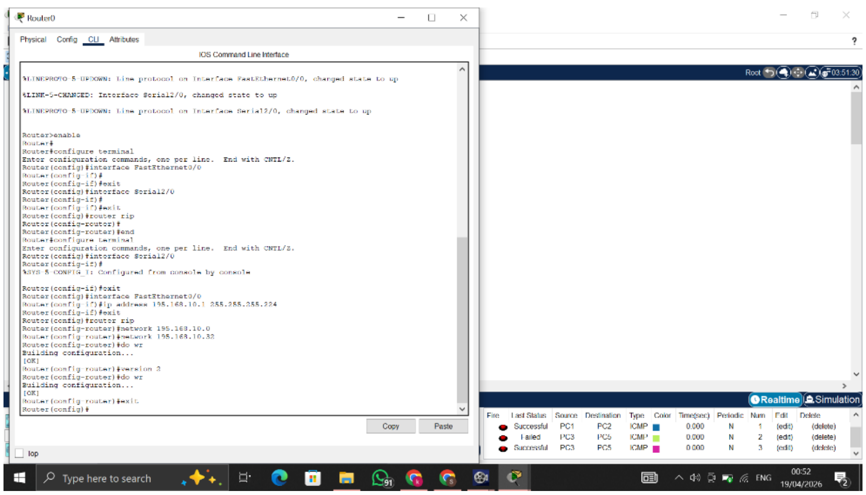
for the Topology given:- 5 net required
$2^3 = 8$ net, 3 bit
255.255.255.224
↳ subnet.

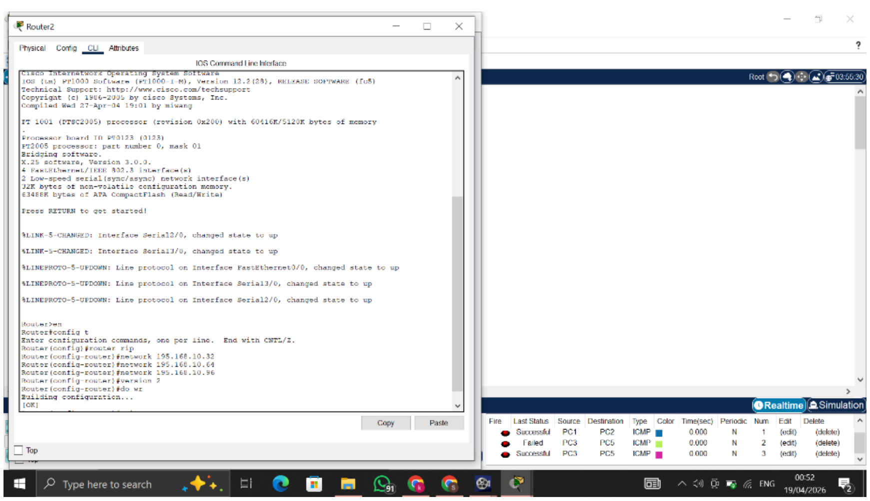
Implementation of the given topology on Cisco:

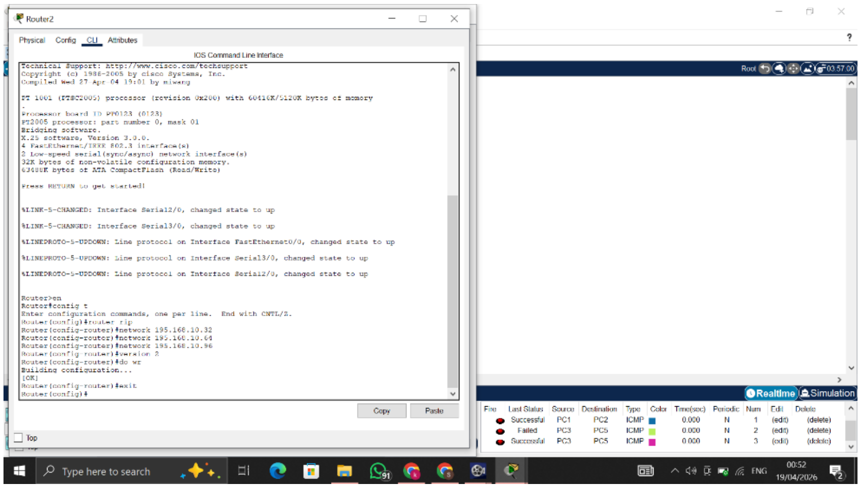


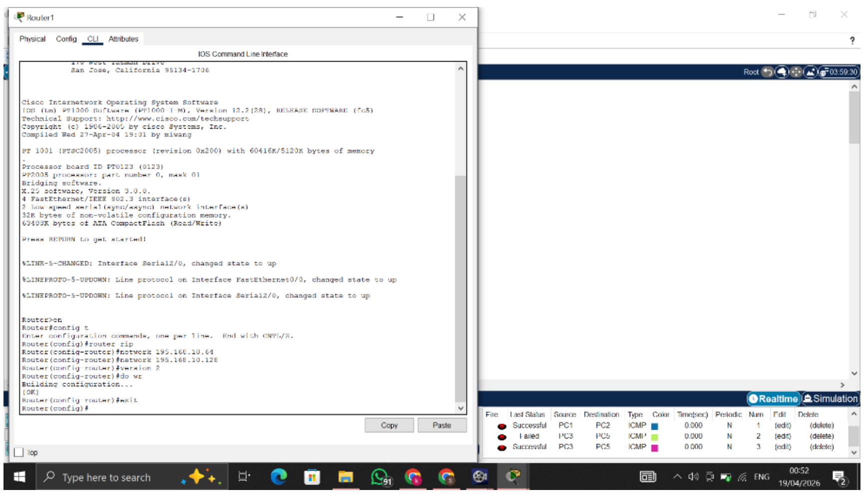


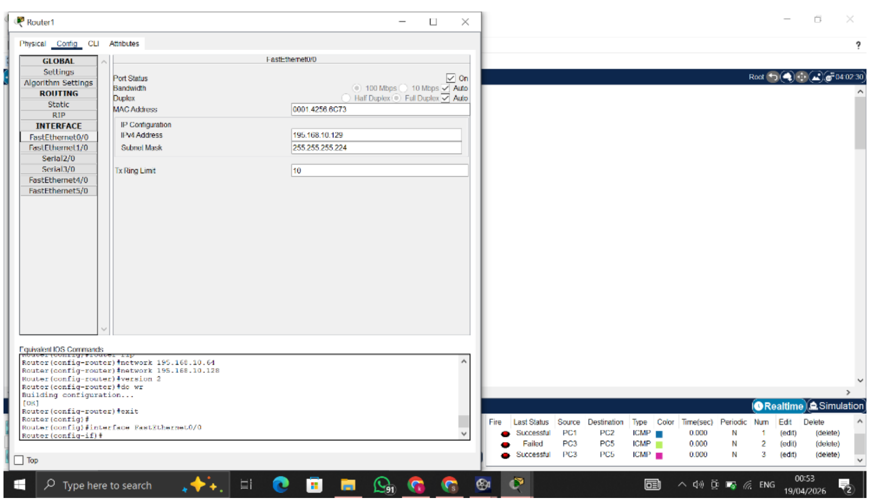

Router2
Physical Config CLI Attributes
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 195.168.10.32
Router(config-router)#network 195.168.10.64
Router(config-router)#network 195.168.10.96
Router(config-router)#version 2
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#
Copy
Paste
Top
Type here to search
Realtime
Simulation

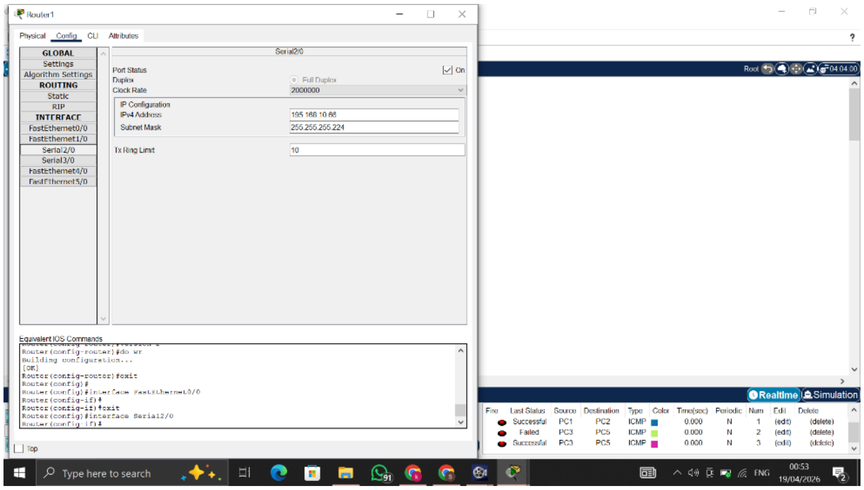

Router0
Physical Config CLI Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 195.168.10.1 255.255.255.224
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 195.168.10.0
Router(config-router)#network 195.168.10.32
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#version 2
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#
Copy
Paste
Top
Type here to search
Realtime
Simulation


Router2
Physical Config CLI Attributes
IOS Command Line Interface
Press RETURN to get started!
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 195.168.10.32
Router(config-router)#network 195.168.10.64
Router(config-router)#network 195.168.10.96
Router(config-router)#version 2
Router(config-router)#do wr
Building configuration...
[OK]
Copy
Paste
Top
Type here to search
Realtime
Simulation

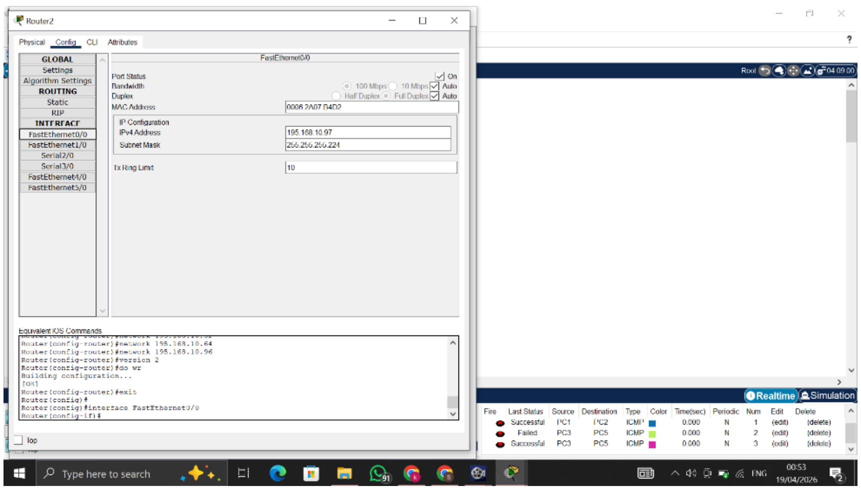

Router2
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINK-5-CHANGED: Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 195.168.10.32
Router(config-router)#network 195.168.10.64
Router(config-router)#network 195.168.10.96
Router(config-router)#version 2
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#
Copy
Paste
Top
Realtime
Simulation
Type here to search


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Cisco Internetwork Operating System Software
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 195.168.10.64
Router(config-router)#network 195.168.10.128
Router(config-router)#version 2
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#
Copy
Paste
Top
Realtime
Simulation
Type here to search

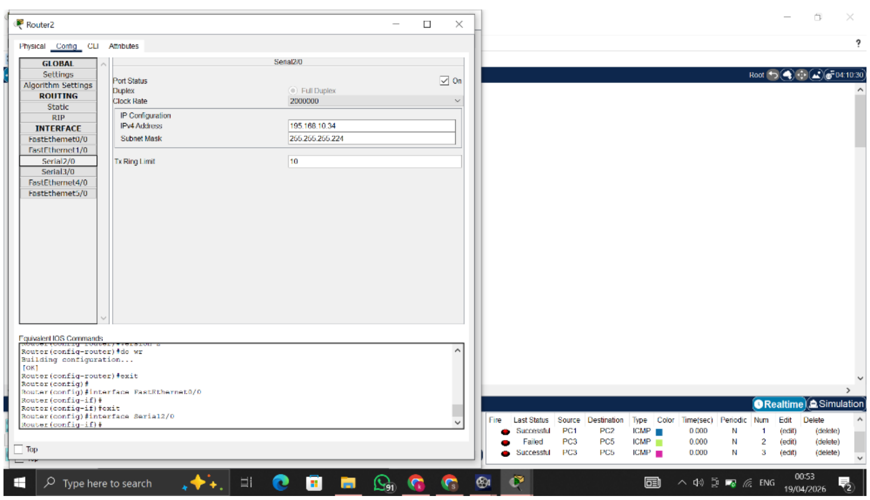

Router1
Physical
Config
CLI
Attributes
FastEthernet0/0
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Port Status
Bandwidth
Duplex
MAC Address
0001.4256.6C73
IP Configuration
IPv4 Address
195.168.10.129
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-router)#network 195.168.10.64
Router(config-router)#network 195.168.10.128
Router(config-router)#version 2
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Top
Realtime
Simulation
Type here to search



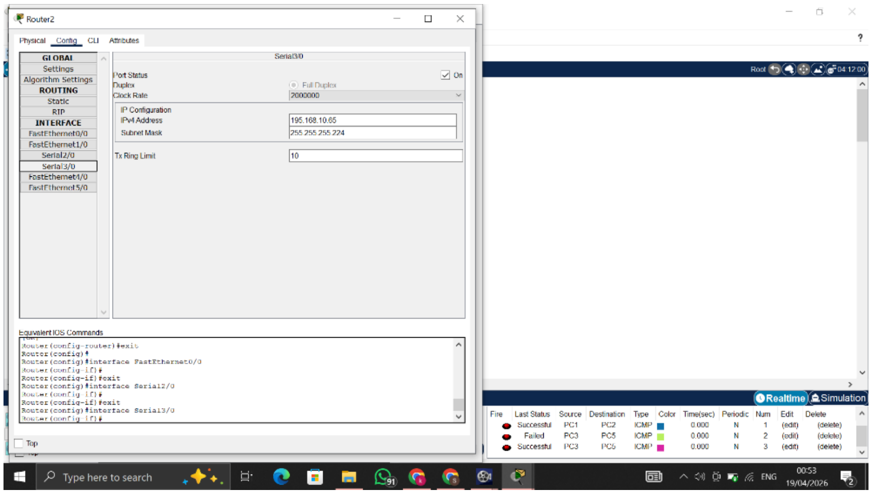
Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial2/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
195.168.10.66
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
Top
Realtime
Simulation
Type here to search

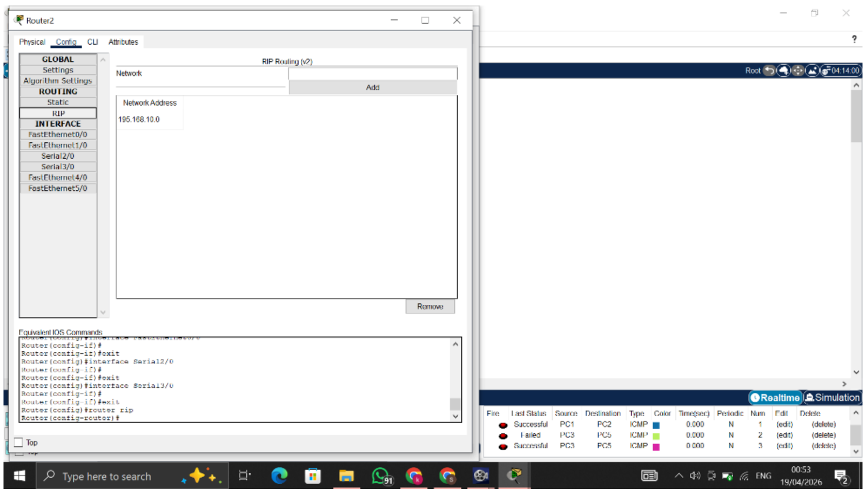


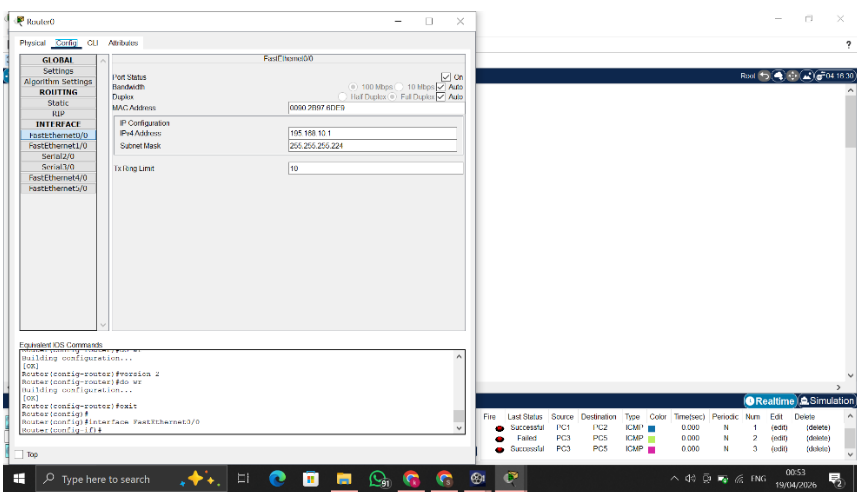
Router2
Physical
Config
CLI
Attributes
FastEthernet0/0
Port Status
On
Bandwidth
Duplex
MAC Address
IP Configuration
IPv4 Address
195.168.10.97
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
Top
Realtime
Simulation
Type here to search

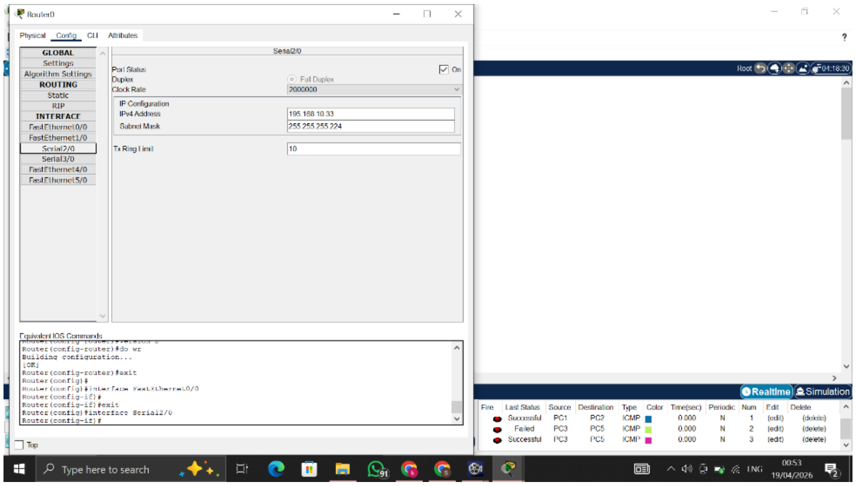


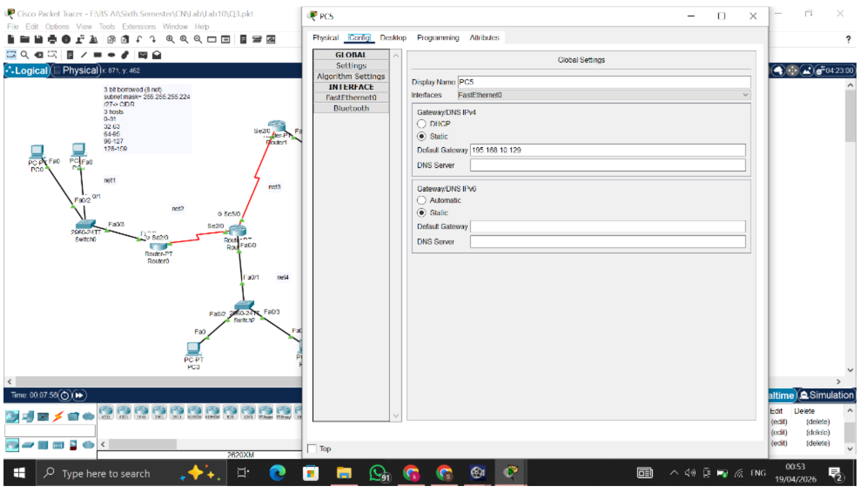
Router2
Physical
Config
CLI
Attributes
Serial2/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
195.168.10.34
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
Top
Realtime
Simulation
Type here to search

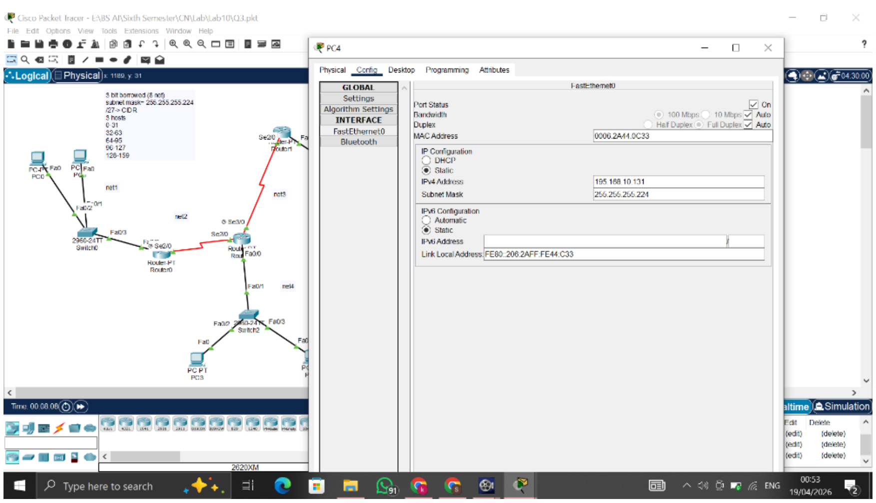

Router2
Physical
Config
CLI
Attributes
Serial3/0
Port Status
Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
195.168.10.65
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
Realtime
Simulation
Type here to search

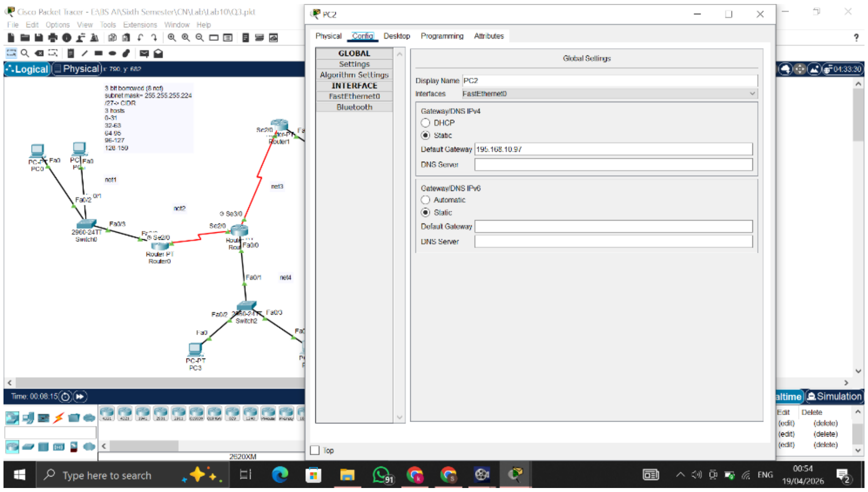

Router2
Physical
Config
CLI
Attributes
RIP Routing (v2)
Network
Add
Network Address
195.168.10.0
Remove
Equivalent IOS Commands
Router(config)#router rip
Router(config-router)#
Realtime
Simulation
Type here to search

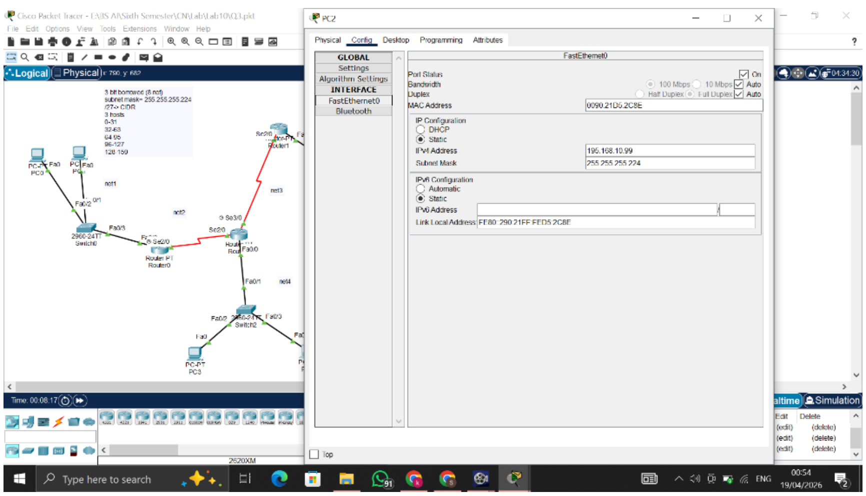

Router0
Physical
Config
CLI
Attributes
FastEthernet0/0
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
IPv4 Address
195.168.10.1
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-router)#version 2
Router(config-router)#do wr
Router(config)#interface FastEthernet0/0
Realtime
Simulation
Type here to search

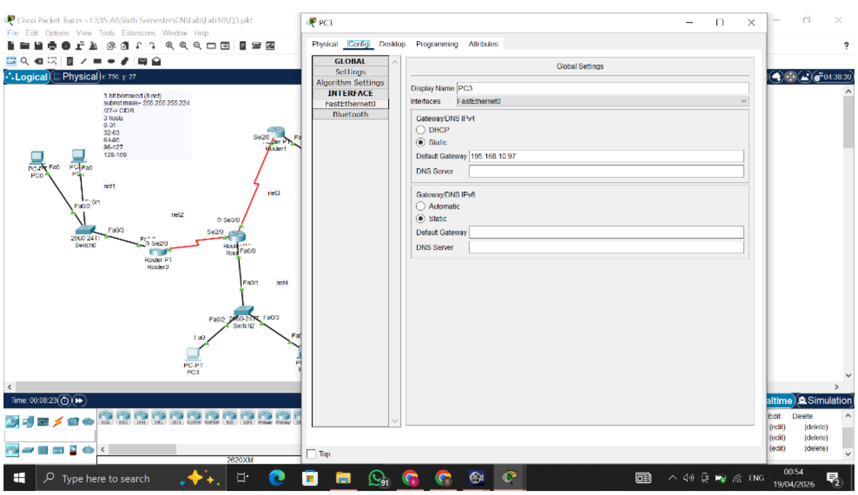

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Serial2/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
195.168.10.33
Subnet Mask
255.255.255.224
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#exit
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
Top
Realtime
Simulation
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Successful
PC1
PC2
ICMP
0.000
N
1
(edit)
(delete)
Failed
PC3
PC5
ICMP
0.000
N
2
(edit)
(delete)
Successful
PC3
PC5
ICMP
0.000
N
3
(edit)
(delete)
Type here to search
ENG
00:53
19/04/2026

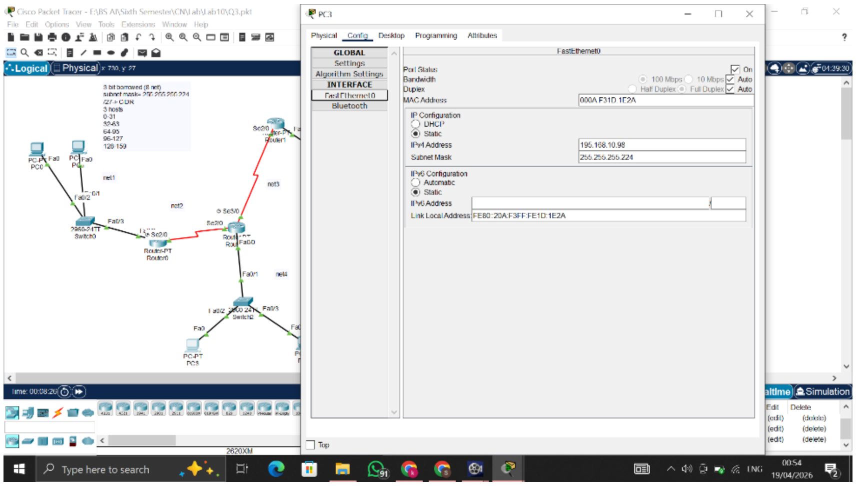

PC5
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC5
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
195.168.10.129
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Logical
Physical
3 bit borrowed (8 net)
subnet mask= 255.255.255.224
/27-> CIDR
5 hosts
0-31
32-63
64-95
96-127
128-159
net1
net2
net3
net4
Switch0
Router0
Switch2
Time: 00:07:56
Type here to search
ENG
00:53
19/04/2026

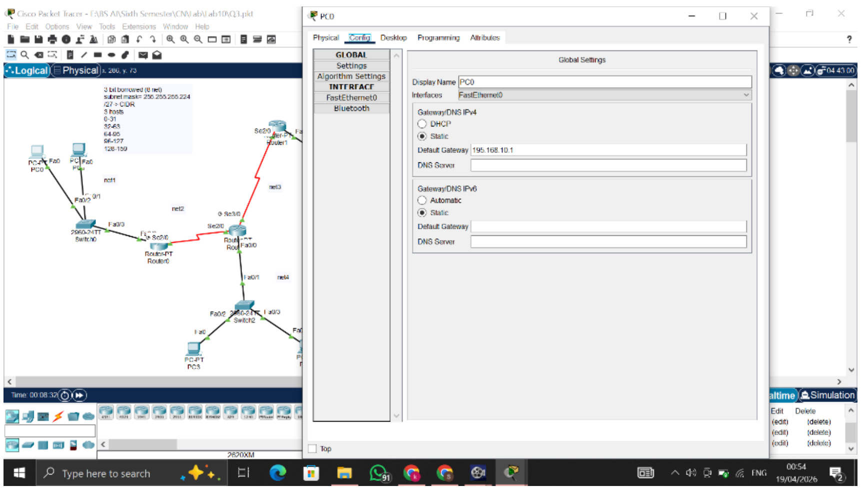

PC4
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0006.2A44.0C33
IP Configuration
DHCP
Static
IPv4 Address
195.168.10.131
Subnet Mask
255.255.255.224
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::206:2AFF:FE44:C33
Logical
Physical
3 bit borrowed (8 net)
subnet mask= 255.255.255.224
/27-> CIDR
5 hosts
0-31
32-63
64-95
96-127
128-159
net1
net2
net3
net4
Switch0
Router0
Switch2
Time: 00:08:06
Type here to search
ENG
00:53
19/04/2026

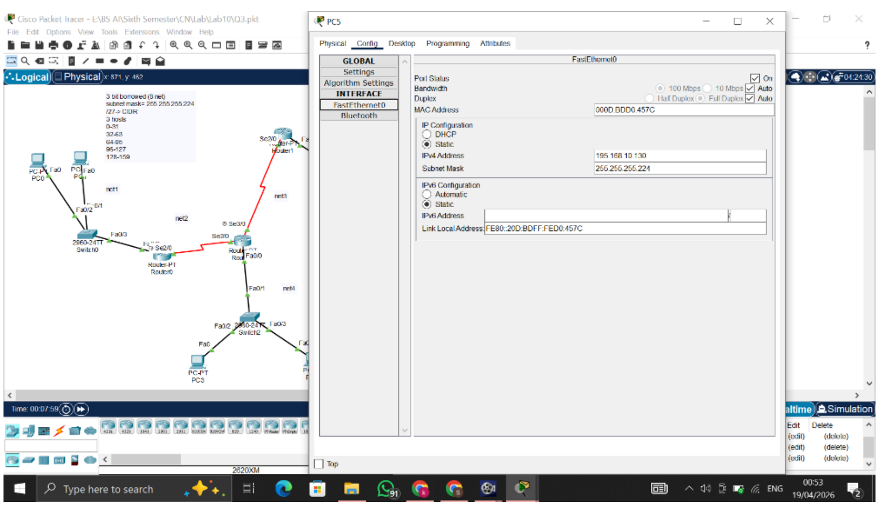

PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC2
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 195.168.10.97
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Type here to search

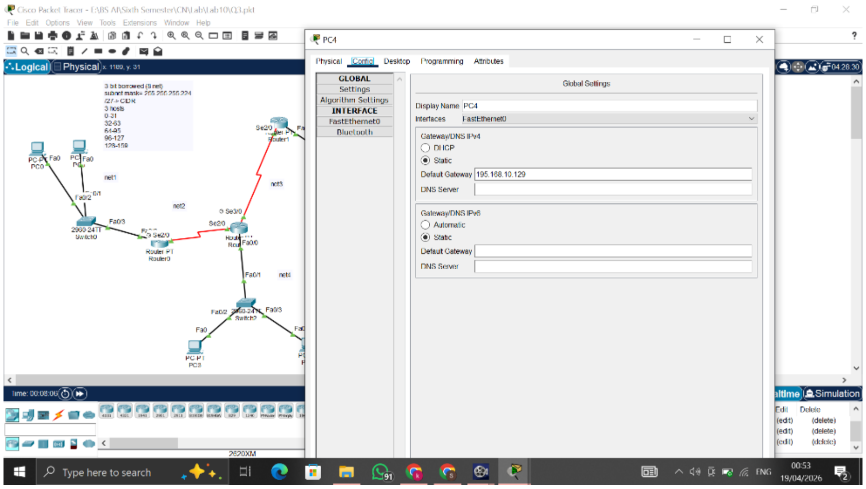

PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address 0090.21D5.2C8E
IP Configuration
DHCP
Static
IPv4 Address 195.168.10.99
Subnet Mask 255.255.255.224
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::290:21FF:FED5:2C8E
Top
Type here to search

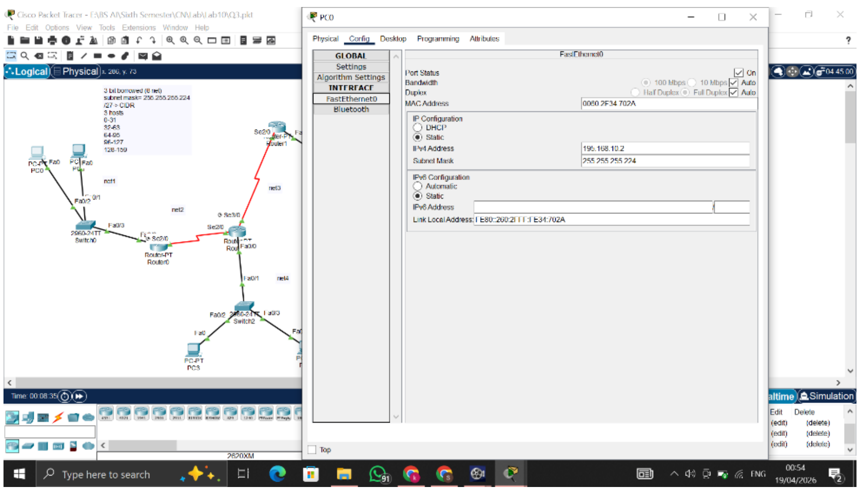

PC3
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC3
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 195.168.10.97
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Type here to search

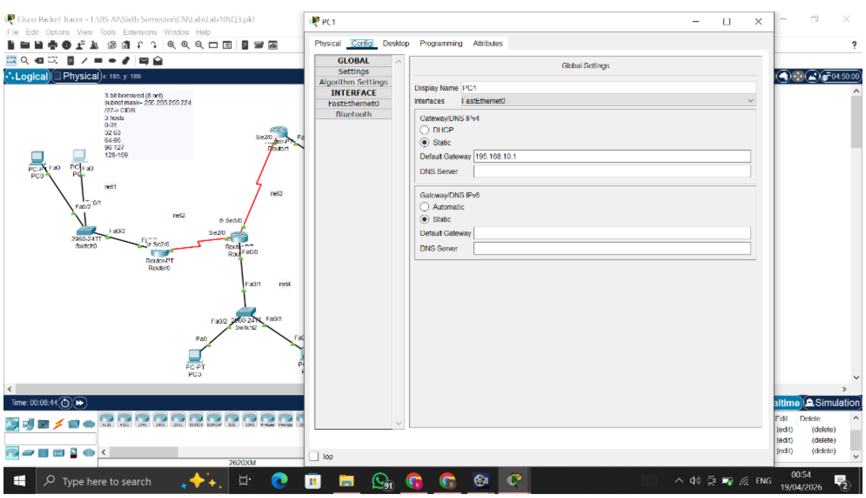

PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
DHCP
Static
IPv4 Address
195.168.10.98
Subnet Mask
255.255.255.224
IPv6 Configuration
Automatic
IPv6 Address
Link Local Address
Top
Type here to search

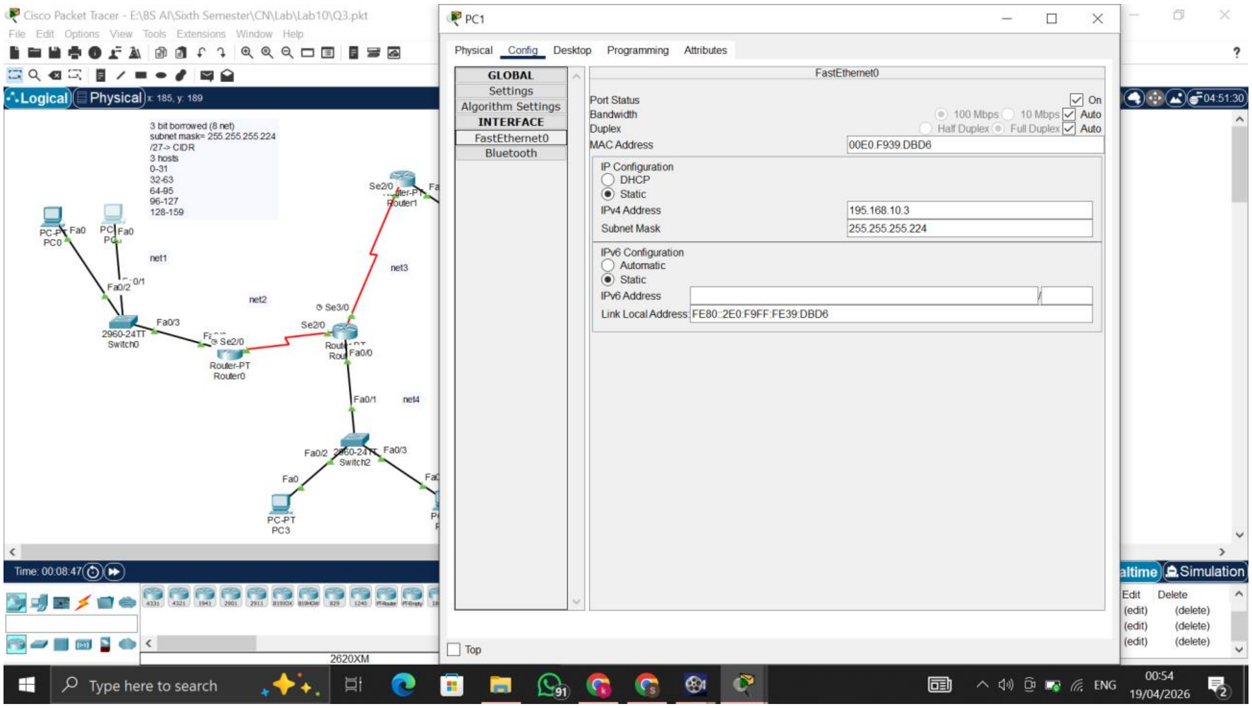

PC0
Physical
Config
Desktop
Programming
Attributes
Global Settings
Display Name
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
195.168.10.1
DNS Server
Gateway/DNS IPv6
Automatic
Top
Type here to search

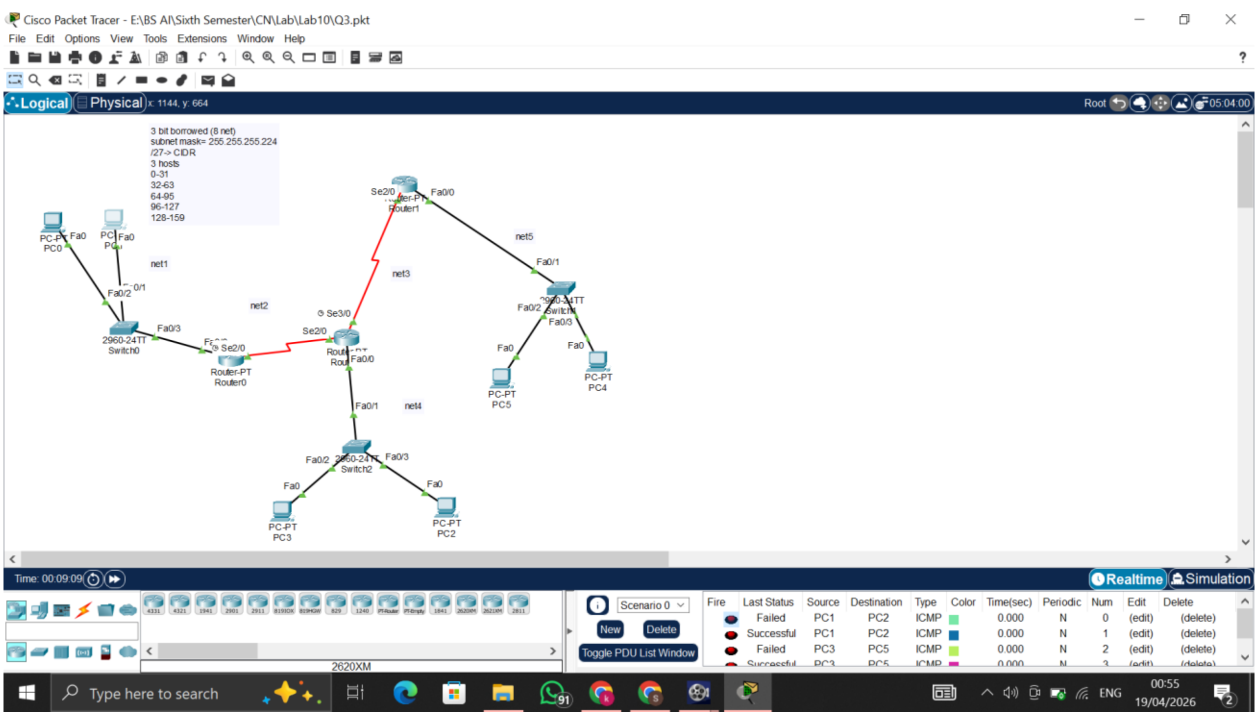

PC5
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
DHCP
Static
IPv4 Address
195.168.10.130
Subnet Mask
255.255.255.224
IPv6 Configuration
Automatic
IPv6 Address
Link Local Address
Top
Type here to search


PC4
Physical
Config
Desktop
Programming
Attributes
Global Settings
Display Name PC4
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 195.168.10.129
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Type here to search


PC0
Physical
Config
Desktop
Programming
Attributes
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
DHCP
Static
IPv4 Address 195.168.10.2
Subnet Mask 255.255.255.224
IPv6 Configuration
Automatic
Static
IPv6 Address
Type here to search


PC1
Physical
Config
Desktop
Programming
Attributes
Global Settings
Display Name PC1
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 195.168.10.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Type here to search


Cisco Packet Tracer - E:\BS AI\Sixth Semester\CN\Lab\Lab10\Q3.pkt
PC1
Physical Config Desktop Programming Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.F939.DBD6
IP Configuration
DHCP
Static
IPv4 Address
195.168.10.3
Subnet Mask
255.255.255.224
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::2E0:F9FF:FE39:DBD6
Top
3 bit borrowed (8 net)
subnet mask= 255.255.255.224
/27-> CIDR
3 hosts
0-31
32-63
64-95
96-127
128-159
net1
net2
net3
net4
Switch0
Router0
Router1
Switch2
PC0
PC3
Time: 00:08:47
Type here to search
00:54
19/04/2026


Cisco Packet Tracer - E:\BS AI\Sixth Semester\CN\Lab\Lab10\Q3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1144, y: 664
Root
05:04:00
3 bit borrowed (8 net)
subnet mask= 255.255.255.224
/27-> CIDR
3 hosts
0-31
32-63
64-95
96-127
128-159
net1
net2
net3
net4
net5
2960-24TT
Switch0
Router-PT
Router0
Router1
Switch2
PC-PT
PC0
PC3
PC2
PC5
PC4
Time: 00:09:09
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
2620XM
Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete
Failed | PC1 | PC2 | ICMP | 0.000 | N | 0 | (edit) | (delete)
Successful | PC1 | PC2 | ICMP | 0.000 | N | 1 | (edit) | (delete)
Failed | PC3 | PC5 | ICMP | 0.000 | N | 2 | (edit) | (delete)
Type here to search
00:55
19/04/2026